

Liouville's Theorem

If a function $f(z)$ is analytic for all finite values of z and is bounded, then it is constant.

Fundamental Theorem of Algebra

Every polynomial of degree n has exactly n -zeros.

$$f(z) = z^n + a_1 z^{n-1} + a_2 z^{n-2} + \dots + a_n = 0$$

$$\Rightarrow f(z) = 0.$$

Morera's Theorem

If $f(z)$ is continuous and $\oint_C f(z) dz = 0$ within and on a closed curve C , then $f(z)$ is analytic.